

1.6 Semester-VI Table and Courses

Sem-6			
Subject	Type	L-T-P	Credit
File organization and Database Systems	DC	3-0-0	3
File organization and Database Systems Lab	DL	0-0-3	2
Applied Computational Methods	DC	3-0-0	3
Applied Computational Methods Lab	DL	0-0-3	2
Depth Elective-III	DE	3-0-0	3
Computer Organization and Architecture	DC	3-0-0	3
Depth Elective-IV	DE	3-0-0	3
Breadth Elective –II	BE	3-0-0	3
Total credits			22

DETAILED SYLLABUS

File Organization and Database Systems L-T-P: 3-0-0 Credit: 3 MAxxxxx

Prerequisites: Programming Languages

Category: Existing

Syllabus:

Introduction: Purpose of database systems, data abstraction and modelling, instances and schemes, database manager, database users and their interactions, data definition and manipulation language, data dictionary, overall system structure. [3 Lectures]

Entity-relationship model: Entities and entity sets, relationships and relationship set, mapping constraints, E-R diagram, primary keys, strong and weak entities, reducing E-R diagrams to tables, trees or graphs, generalization and specialization, aggregation. [5 Lectures]

Brief Introduction to hierarchical and network model: Data description and tree structure diagram for hierarchical model, retrieval and update facilities, limitations; Database task group (DBTG) model, record and set constructs retrieval and update facilities, limitations. [6 Lectures]

Relational model: Structure of a relational database, operation on relations, relational algebra, tuple and domain relational calculus, salient feature of a query language. Structured query language: Description an actual RDBMS and SQL. [4 Lectures]

Normalization: Pitfalls in RDBMS, importance of normalization, functional, multivalued and join dependencies, INF to 5NF, limitations of RDBMS. [3 Lectures]

Database tuning: Index selection and clustering, tuning of conceptual schema, denormalization, tuning queries and views. [3 Lectures]

Query optimization: Importance of query processing, equivalence of queries, cost estimation for processing a query, general strategies, birelational and multi- relational join algorithms, algebraic manipulation. [6 Lectures]

Crash recovery: Failure classification, transactions, log maintenance, check point implementation, shadow paging, example of an actual implementation. [4 Lectures]

Concurrency Control in RDBMS: Testing for serializability, lock based and timestamp-based protocols; Deadlock detection and Recovery. [2 Lectures]

References:

1. H. F. Korth and A. Silberschatz: Database System Concepts, McGraw Hill, New Delhi, 1997.
2. R. A. Elmasri and S. B. Navathe: Fundamentals of Database Systems, 3rd ed., Addison-Wesley, 1998.
3. R. Ramakrishnan: Database Management Systems, 2nd ed., McGraw Hill, New York, 1999.

Teachers: Prof. A. Goswami, Prof. P. Kumar

File Organization and Database Systems Lab L-T-P: 0-0-3 Credit: 2 MAxxxxx

Prerequisites: File Organization and Database Systems

Category: Existing

Syllabus:

Conceptual Designing using ER Diagrams (Identifying entities, attributes, keys and relationships between entities, cardinalities, generalization, specialization etc.). Converting ER Model to Relational Model (Represent entities and relationships in Tabular form, represent attributes as columns, identifying keys). Normalization -To remove the redundancies and anomalies in the above relational tables, Normalize up to Third Normal Form.

Creation of a database and writing SQL queries to retrieve information from the database. Performing Insertion, Deletion, Modifying, Altering, Updating and Viewing records based on conditions. Creation of Views, Synonyms, Sequence, Indexes, Save point.

Creating an Employee database to set various constraints. Creating relationship between the databases.

Study of PL/SQL block. Write a PL/SQL block to satisfy some conditions by accepting input from the user including exception handling. Creation of Procedures. Creation of database triggers and functions.

Mini project (Application Development using Oracle/ Mysql) on :

- a Timetable Management System
- b Hall Management System
- c Web Based User Identification System
- d Hospital Management System
- e Railway Reservation System
- f Inventory Control System

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Applied Computational Methods L-T-P: 3-0-0 Credit: 3 MAxxxxx

Prerequisite: NA

Category: Existing

Syllabus

Algorithms for the solution of linear systems of equations – Direct Methods, LU Decomposition, Thomas Algorithm, Block matrix; Iterative methods, Basic theory of linear fixed-point iteration; Successive overrelaxation (SOR); Conjugate gradients, Iterative methods for Block matrix **[8 lecture]**

Eigenvalue and vector computations including the power method, QR iteration and Lanczos iteration **[6 lecture]**